



Stereum

STEREUM PAY INTEGRATION GUIDE

*@Code:ST-SIS-0006

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* @Versión: 2.4

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* @Confidential Level : **CONFIDENTIAL**

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CHANGE CONTROL TABLE				
Date	Description	Reviewer	Approval	Version
16/04/2025	Preparation of the document	Ing. Ricardo Laredo		1.0
09/05/2025	Update on request and response	Ing. Ricardo Laredo		2.0
26/05/2025	Added: • Definition of the API for canceling a charge. • Added optional metadata for the charge API. • Included the final URL for the development environment.	Ing. Ricardo Laredo		2.1
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28/06/2025	Optional fields have been added to the webhook body, as follows: • amount_received - This field contains the value of the amount received. • network	Ing. Ricardo Laredo		2.3
31/06/2025	Adaptation of concepts and adjustment of fields	Ing. Ricardo Laredo		2.4

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GLOSSARY

This document uses terms that, if not interpreted correctly, could fail to fulfill their purpose, which is why the following glossary has been prepared with the appropriate meaning.

GLOSSARY	
WORD	CONCEPT
Connect Timeout	In a client-server socket or HTTP connection, the Connect Timeout is the maximum time to wait before establishing a connection. After this time, the client no longer attempts to establish a connection. When a Connect Timeout error occurs, the request cannot be sent to the server, so there should be no need to verify processing.
Read Timeout	In an HTTP connection between a client and a server, the Read Timeout is the maximum amount of time the client waits for a response from the server. After this time, the client generates a Read Timeout exception. When a Read Timeout error is generated, there's a chance the server has received and processed the request, so it's recommended to perform a check.

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TLS	The TLS (Transport Layer Security) protocol is simply an updated, more secure version of SSL. The supported versions are 1.2 and 1.3.
Cipher-suite	Set of encryption algorithms that help secure the network connection.
HMAC	HMAC, which stands for “Hash-based Message Authentication Code,” is an authentication mechanism that uses a hash function along with a secret key to generate an authentication code that is appended to the transmitted message.
SHA-256	Algorithm for generating a hash of any string whose length and value will always be the same for any text. It's like obtaining a fingerprint of a text.
API KEY	The API KEY is a unique value that identifies a system. Stereum Pay allows you to generate multiple API KEYS for as many systems as you want, and you can generate them from the administration console.
Console	At Stereum Pay, we call the web application that business users access to manage and process their payments the console. https://console.stereum.tech

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INTRODUCTION

The purpose of this document is to specify the integration between the business core and Stereum-Pay via API.

Stereum Pay is a payment gateway that allows businesses to receive payments in USDT, USDC, and bolivianos via Simple QR codes provided by Bolivian financial institutions.

INITIAL SETUP

ACCOUNT CREATION

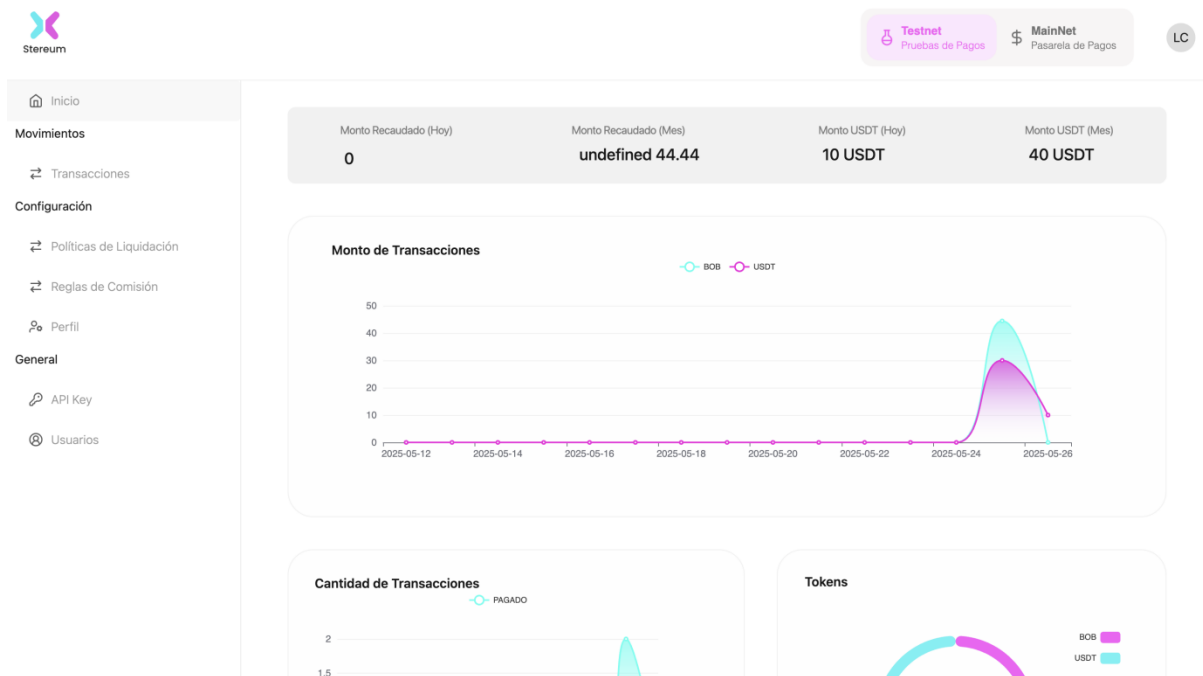
To create a Stereum Pay account, visit <https://console.stereum.tech> and create an account using your corporate email address. Your company information will also be requested during the account creation process.

API KEY GENERATION

In the Stereum Pay console, you'll need to generate an API key by following these steps:

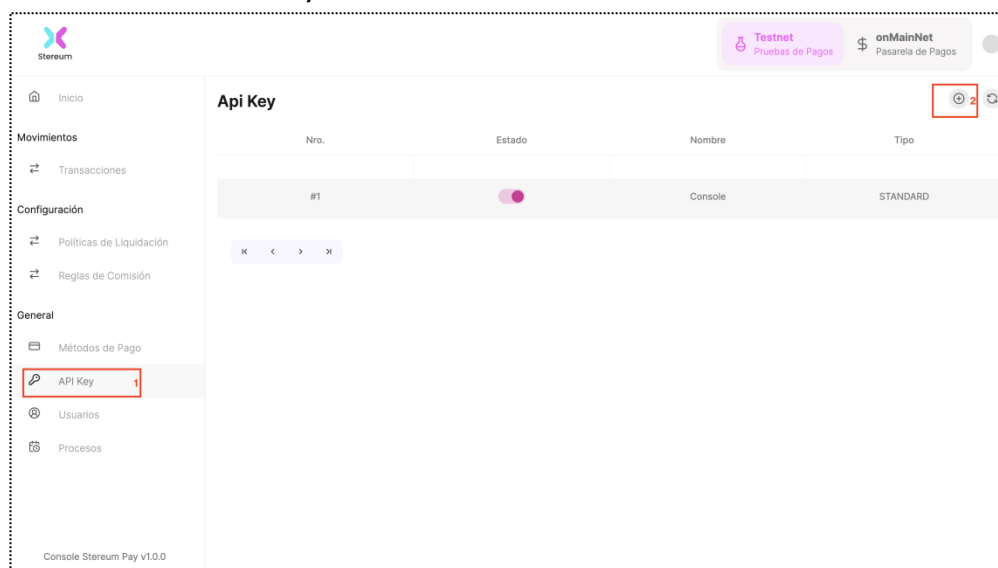
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Step 1.- Log in to the Stereum Pay admin console



Dashboard de la consola de administración de Stereum Pay

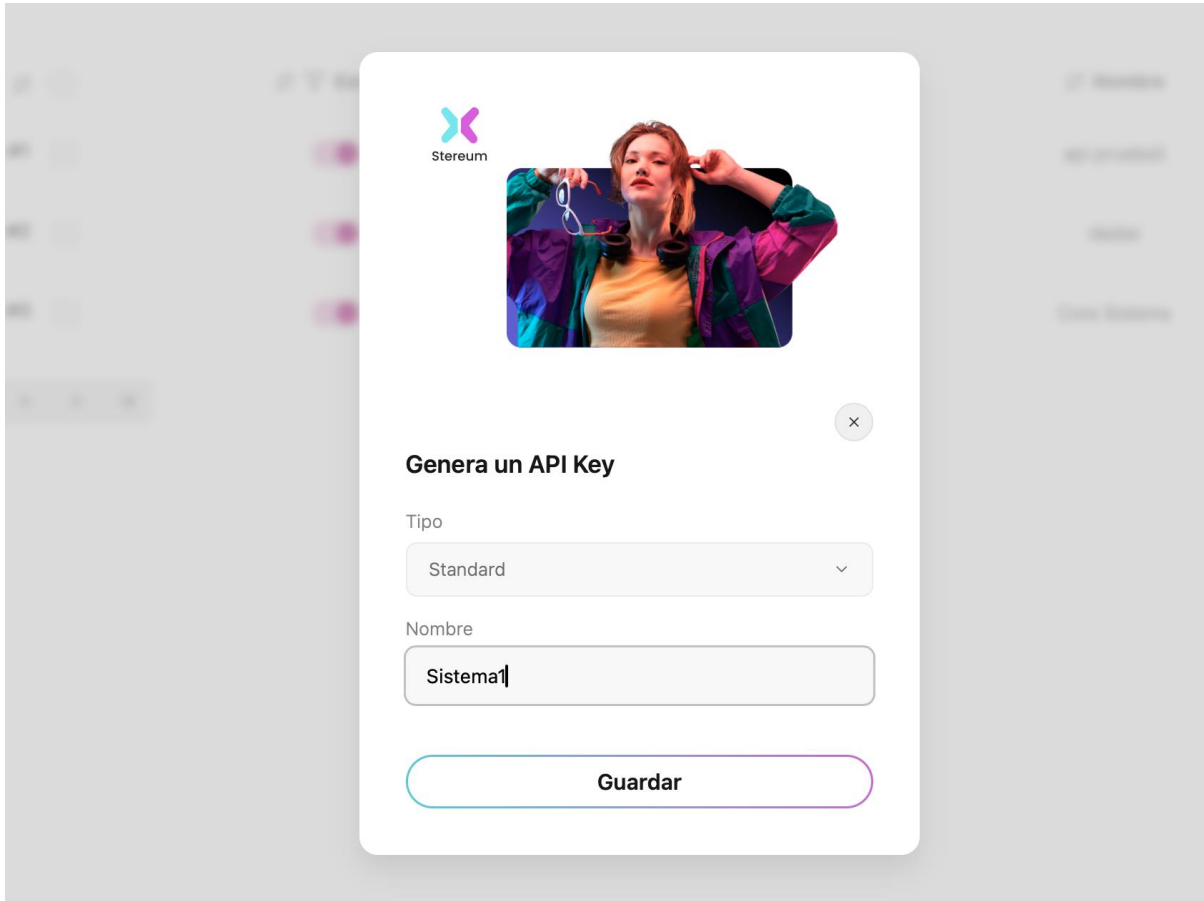
Step 2.- Go to the “API Key” section and click the button to create a new one



API Key interface

Step 3.- Enter the name of your system and press the “Save” button.

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Genera un API Key

Tipo

Standard

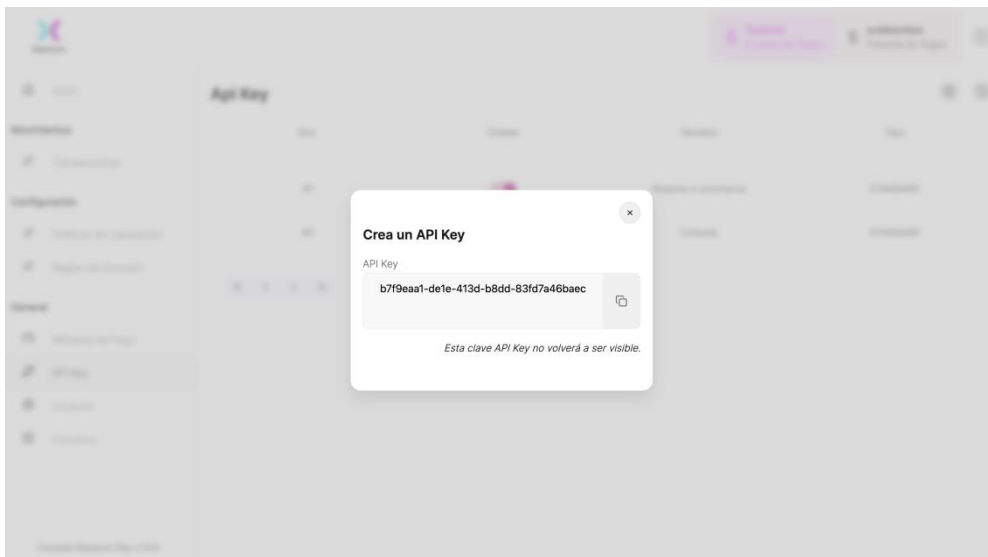
Nombre

Sistema

Guardar

API Key interface

Step 4.- Save the generated API Key safely.



Crea un API Key

API Key

b7f9eaa1-de1e-413d-b8dd-83fd7a46baec

Esta clave API Key no volverá a ser visible.

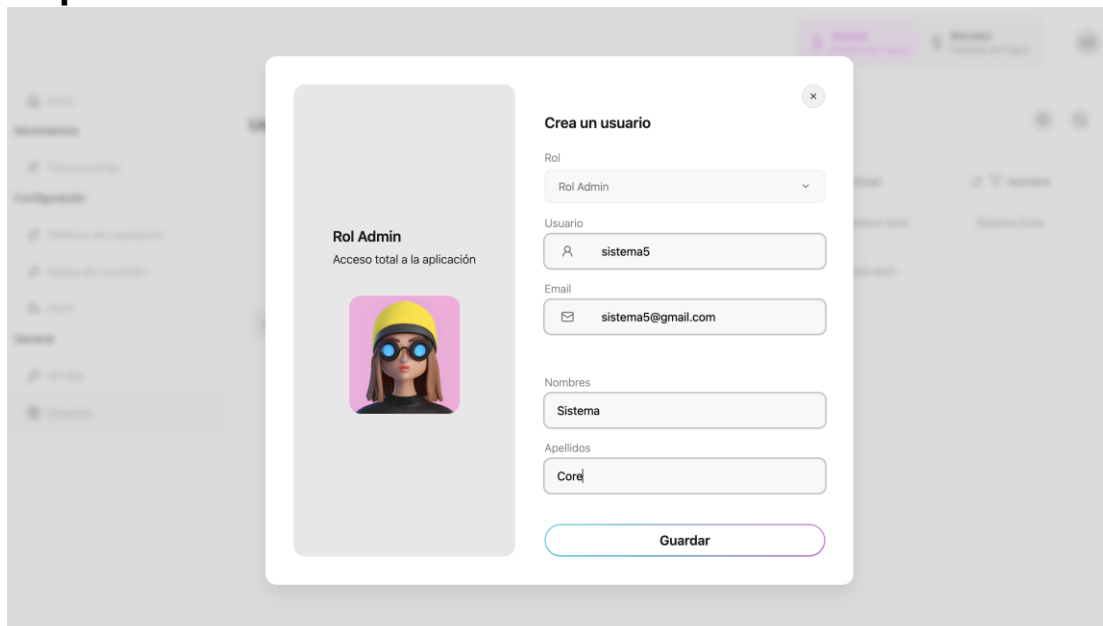
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Interface showing the created API Key.

USER GENERATION FOR AUTHENTICATION

Stereum Pay uses a username and password for authentication. You can generate this username and password in the Administration console by following these steps:

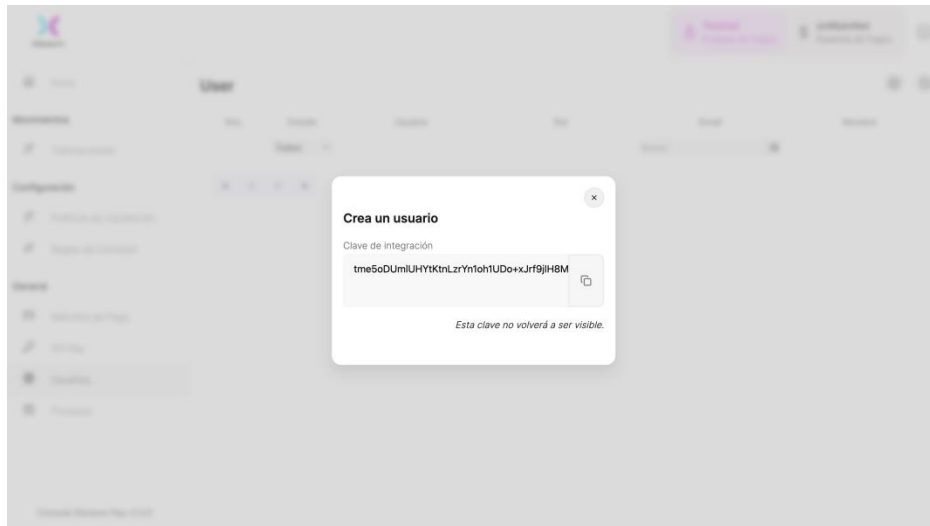
Step 1. Go to the user interface and create a new user with an API role.



Interfaz para crear usuario.

Step 2.- Interface that shows the password of the created user and that must be used to obtain the authentication token.

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Interface that displays the user's password.

PUBLIC KEY GENERATION

To ensure the confidentiality of your authentication data, your password must be encrypted using a public key and the RSA asymmetric encryption algorithm.

The certificate will be provided by the Stereum team via email.

API INTEGRATION

Stereum Pay has the following APIs for integration.

URL Base: **https://api.stereum.tech**

API	ENDPOINT
API to obtain an authentication TOKEN	POST /api/v1/auth/token

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API to create a collection charge	POST /api/v1/transactions/create-charge
API to obtain the status of a collection	GET /api/v1/transactions/{{TRANSACTION_ID}}/verify
API to convert an amount in BOB to an AMOUNT in USDT or USDC	GET /api/v1/currency/convert?country=BO&from=BOB&to=USDT &amount=100
API to cancel a charge	POST /api/v1/transactions/{{TRANSACTION_ID}}/cancel



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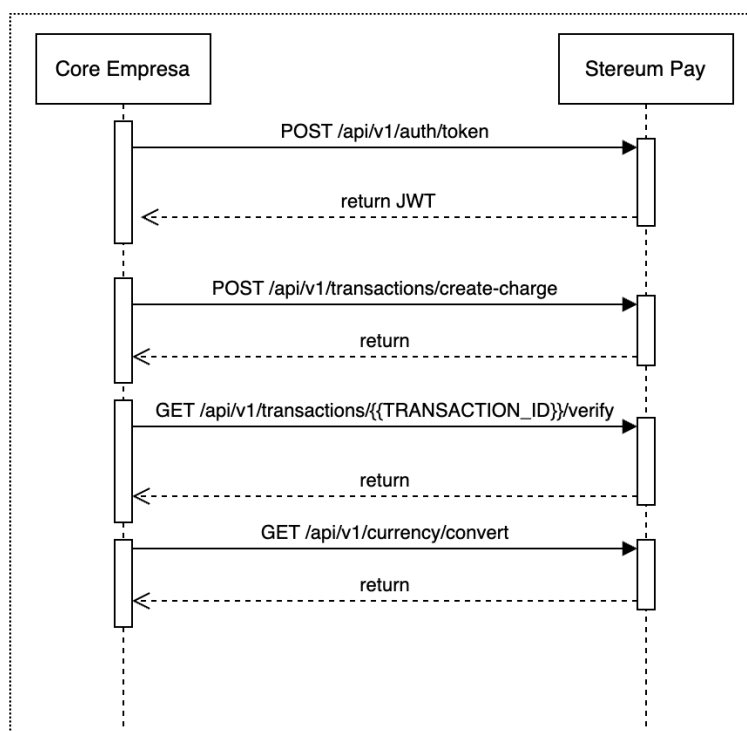


Figura 2. Stereum's APIs

OBTAINING TOKEN

This API allows you to obtain a JWT (JSON Web Token) valid for 8 hours; the JWT must be sent in the Auth header of the other APIs, as specified below.

Aware that the TLS v1.2, v1.3, and cipher suite protocols can be compromised at some point, at Stereum, we decided that sensitive data such as passwords should travel over the network encrypted with the RSA algorithm. You can download the public key required for encryption by following the steps mentioned in this section.

End point	POST /api/v1/auth/token
Request	{

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	<pre>"username": "ecommerce", "password": "[PASSWORD ENCRYPTADO]" }</pre>
Response 201	<pre>{ "access_token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOiJlY29tbWVhY2UiLCJqdGkiOiJlZTkwOGUwZi0yNTAzLTU0OTUyYTY5Yy0xYWVlYTc4MzIxOGliLCJpYXQiOiJlZTkwOGUwZi0yNTAzLTU0OTUyYTY5Yy0xYWVlYTc4MzIxOGliLCJlbWFpbF92ZXJpZmllZC16dHJ1ZX0.0SquVegPTC_e5jte2brQXmHWprvzpc9-kj42KDlmfsl", "id_token": "eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOiJlY29tbWVhY2UiLCJqdGkiOiJlZTkwOGUwZi0yNTAzLTU0OTUyYTY5Yy0xYWVlYTc4MzIxOGliLCJpYXQiOiJlZTkwOGUwZi0yNTAzLTU0OTUyYTY5Yy0xYWVlYTc4MzIxOGliLCJlbWFpbF92ZXJpZmllZC16dHJ1ZX0.0SquVegPTC_e5jte2brQXmHWprvzpc9-kj42KDlmfsl", "refresh_token": "z9c1AdmDPPHqqmhFCpmZtqu49UdBLJ2vzjh62jNFiE8=", "token_type": "bearer", "expires_in": 480, "expires_at": 1746882856424, "provider_id": "stereum", "sign_in_provider": "stereum", "user": { "id": "3e918a0f-2503-4b95-a69c-1ab1a783218b", "status": "ACTIVO", "user_type": "EMPRESA", "created_at": "2025-05-10T01:00:08.387-04:00", "updated_at": "2025-05-10T01:00:08.387-04:00", "aud": "authenticated", "role": "ROLE_API", "email": "ecommerce@stereum.tech", "username": "ecommerce", "name": "Sistema ecommerce", "email_confirmed_at": null, "phone": null, "confirmation_sent_at": "2025-05-10T01:00:08.387-04:00", "confirmed_at": null, "last_sign_in_at": "2025-05-10T01:14:16.396-04:00", "recovery_sent_at": null, "email_change_sent_at": null, "new_email": null, "new_phone": null, "phone_confirmed_at": null, "is_anonymous": false, "kyc_status": null, "pin_created": true, "password_created": true, "provider_id": "stereum", } }</pre>

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	<pre> "sign_in_provider": "stereum", "user_metadata": { "sub": "3e918a0f-2503-4b95-a69c-1ab1a783218b", "country": null, "email_verified": true, "birthdate": null, "document_number": null, "second_surname": "ecommerce", "_2fa_biometrics": false, "first_surname": "Sistema", "locale_code": "en-US", "user_type": "EMPRESA", "phone_verified": false, "terms_conditions": "https://stereum.tech/info/", "name": "Sistema ecommerce", "phone_number": null, "exp": null, "email": "ecommerce@stereum.tech", "username": "ecommerce", "document_type": null, "status": "ACTIVO", "privacy_policies": "https://stereum.tech/info/" }, "session_id": "94f1c8e8-c2bb-4ab4-9636-7538621173de" </pre>
Response 500	<pre> { "timestamp": "2025-04-21T16:14:34.545-04:00", "status": 500, "error": "Internal Server Error", "message": "Usuario o contraseña incorrecta", "path": "/api/v1/auth/token" } </pre>

CREATE COLLECTION CHARGE

This API allows you to create a collection charge in USDT, USDC, or BOB and must be invoked from a POS (Point of Sale), ERP, or any other system that processes the sale. The expected flow, but not limited to, is as follows:

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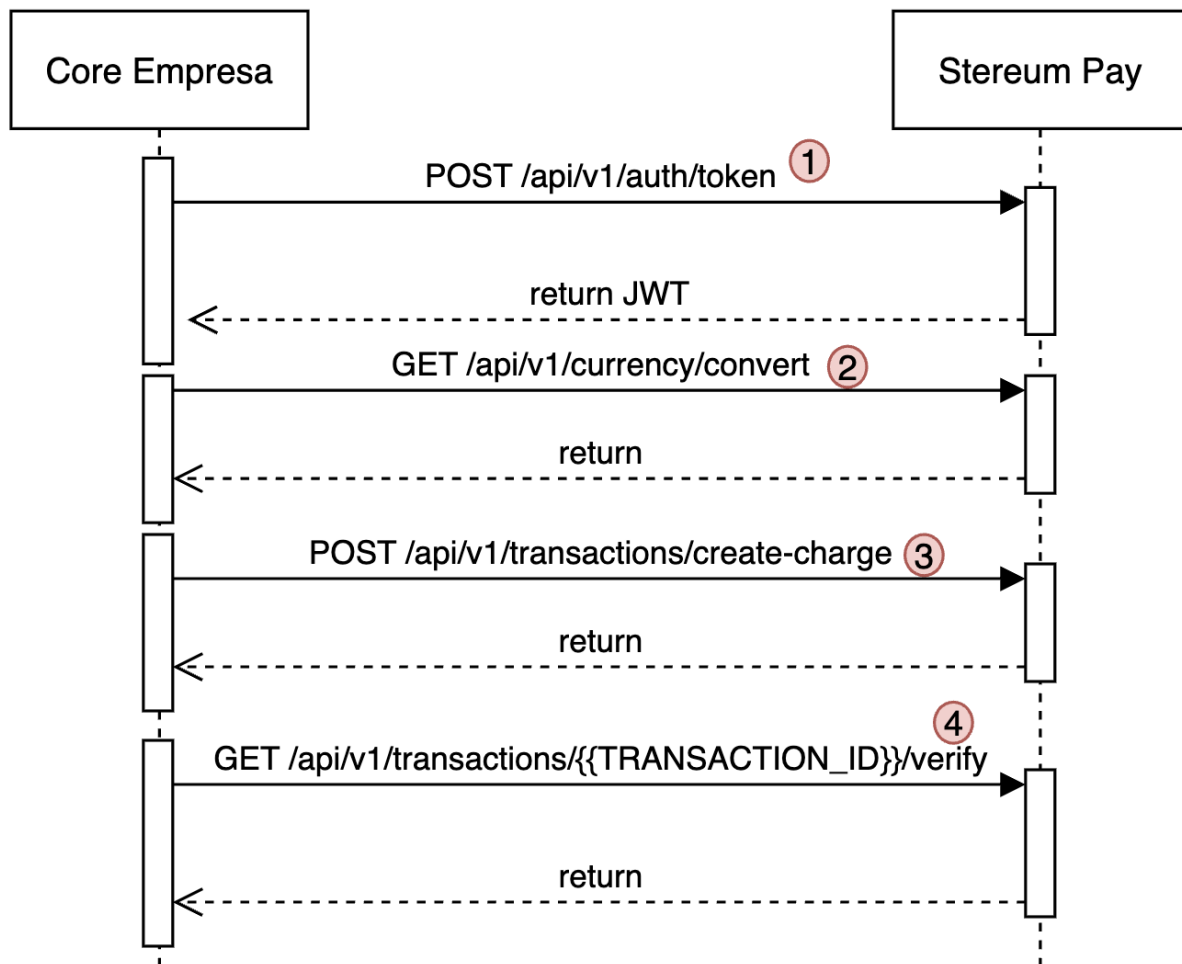


Figure 3. Flow to create a collection charge

1. First, you must call the API to obtain the token using the credentials provided by the administration console. The JWT lasts 8 hours, so you can reuse it to avoid calling this API for each transaction.
2. **Optionally**, you can call the API to convert an amount in bolivianos to USDT.
3. With the amount already defined, you can create the collection charge following the steps and criteria explained in this section.
4. **Optionally**, every so often, or when the customer indicates they have already paid, you can call this API to verify if the payment has indeed been made.

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Response 400	<pre>{ "timestamp": "2025-04-21T17:15:37.488-04:00", "status": 400, "error": "Bad Request", "message": "No hay billteras disponibles, por favor intente en unos minutos", "path": "/api/v1/transactions/create-charge" }</pre>
Response 500	<pre>{ "timestamp": "2025-04-21T17:15:37.488-04:00", "status": 500, "error": "Internal error", "message": "Error interno", "path": "/api/v1/transactions/create-charge" }</pre>

Description of the request fields

country	Country code from which the API is invoked. Default value is BO
amount	- In this field, you must enter the Amount to be Collected and must consider the following validations: - It must be a positive value and greater than zero. - It must contain at most two decimal places. - The decimal separator is a period (.). - To collect amounts in BOB, the maximum amount allowed is 69,600.00
network	In this field, you must enter the network or bank where you want to collect. The values are based on the company's configuration in the administration console. Possible values are: POLYGON

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	• CSL This field is optional; please contact Stereum to delete it.
currency	In this field, you must enter the currency you are charging in; possible values are: • BOB • USDT • USDC The POLYGON network supports USDT and USDC, and the CSL network supports BOB.
Idempotency_key	This value is very important and should, if possible, be the unique identifier of the transaction linked to this charge. The field supports up to 50 characters, and Stereum controls its uniqueness. It is suggested that it be a value generated by UUID (Universally Unique Identifier ID) or GUID (Global Unique Identifier).
charge_reason	In this field, you must provide a summary or job description. For transactions in the financial system, the field must be less than 60 characters long, and for crypto transactions, it must be 200 characters long.
callback	<i>This field is optional and is intended for use when using the payment link.</i> <i>When the customer who received the payment link opens it in a browser and pays, the payment link application will redirect to this link.</i>

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reservation_validity_time	This field is optional and is used to define the QR code's lifetime, expressed in minutes. The QR code's validity period will be 10 minutes by default.
Customer --- Object	
name	Customer name. This field is required for payments in Bolivian currency and supports a length of 60 characters.
lastname	Customer's last name. This field is required for payments in Bolivian currency and supports a length of 60 characters.
document_number	Customer ID number. This field is required for payments in Bolivian currency and supports a length of 20 characters.
email	Customer email address (optional)
phone	Customer phone number (optional)
address	Customer address (optional)
city	City where the customer lives (optional)
country	Two-digit country code of where the customer lives (optional)
state	State or department where the client lives (optional)
Zip_code	Zone identifier or zip code (optional)

Description of the response fields

amount	Returns the amount or value for which the charge was generated.
network	Returns the network on which the charge was generated
currency	Currency in which the charge was generated.

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transaction_status	This field returns the status of the transaction. If the charge was generated correctly, its value will always be PENDING .
id	36-character unique identifier in Stereum Pay.
qr_base64	In this field the QR code is returned in base 64 ready to be displayed by a component.
payment_link	This field returns the payment link, in case you want to share it with the customer or display it in an iframe of a web application or webconnect of a mobile application.
collecting_account	For charges with USDT or USDC currency, this field will return a wallet address to which the customer must transfer the charge amount.
on_main_net	Indicates whether the transaction was created on testnet or mainnet.
expiration_time	Date and time the position expires

CONVERSION API

The BOB to USDT or USDC currency conversion API is optional and is based on the exchange rate published by BCB on its website and other sites.

Construction of the request:

End point	GET /api/v1/currency/convert
Request	The values passed in the URL parameters are as follows: - country=BO - from=BOB - to=USDT - amount=100

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	The amount must be greater than zero and use a period (.) as the decimal separator. Example: <code>/api/v1/currency/convert?country=BO&from=BOB&to=USDT&amount=100</code>
Response 200	<pre>{ "country": "BO", "from_currency": "BOB", "to_currency": "USDT", "amount": 100.00, "converted_amount": 7.67, "exchange_rate": 13.0400000000 }</pre>
Response 400	<pre>{ "timestamp": "2025-04-21T18:38:12.080-04:00", "status": 400, "error": "Bad Request", "message": "No se encontró un tipo de cambio con los criterios ingresados", "path": "/api/v1/currency/convert" }</pre>
Response 500	<pre>{ "timestamp": "2025-04-21T18:38:12.080-04:00", "status": 500, "error": "Internal error", "message": "Error interno", "path": "/api/v1/currency/convert" }</pre>

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API TO CHECK CHARGE STATUS (TRANSACTION)

Stereum offers a webhook for real-time notifications when a payment is made, but if a webhook isn't available, there's this API that returns the current status of the charge or transaction.

Construction of the request:

End point	GET /api/v1/transactions/{{TRANSACTION_ID}}/verify
Request	The {{TRANSACTION_ID}} must be replaced by the identifier returned by Stereum when creating the charge. <i>Example.</i> /api/v1/transactions/{{TRANSACTION_ID}}/verify
Response 200	<pre>{ "id": "e7915e24-6ef1-4a81-8b67-70f36d0423ef", "amount": 1.00, "currency": "USDT", "country": "BO", "status": "PENDIENTE", "fee": 0, "created_date": 1746854277163, "status_description": "Pendiente", "on_main_net": false, "idempotency_key": "8f1cfa93-dd68-4c33-b419-959bd7b9c45d", "expiration_time": 1750169018821 }</pre>
Response 400	<pre>{ "timestamp": "2025-04-21T21:49:25.922-04:00", "status": 400, "error": "Bad Request", "message": "Transacción no encontrada", "path": "/api/v1/transactions/1468c46c-cd35-4ebc-b929-fb8127545fa6s/verify" }</pre>

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	}
Response 500	<pre>{ "timestamp": "2025-04-21T21:49:25.922-04:00", "status": 500, "error": "Internal Error", "message": "Error interno", "path": "/api/v1/transactions/1468c46c-cd35-4ebc-b929-fb8127545fa6s/verify" }</pre>

Note.

The value of the `created_date` field is the date and time the record was created, expressed in milliseconds.

API TO CANCEL A CHARGE (TRANSACTION)

This API is used to cancel the collection request.

Construction of the request:

End point	POST /api/v1/transactions/{{TRANSACTION_ID}}/cancel
Request	The {{TRANSACTION_ID}} It must be replaced by the identifier returned by Stereum when creating the charge. <i>Example.</i> <pre>/api/v1/transactions/e7915e24-6ef1-4a81-8b67-70f36d0423ef/cancel</pre>
Response 200	<pre>{ "id": "835fab3d-a80e-4581-ae81-9c82a5f90d81", "amount": 100.00, "currency": "BOB", }</pre>

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	<pre> "country": "BO", "status": "CANCELADA", "fee": 0, "created_date": 1753996654907, "amount_received": 0, "status_description": "Cancelada", "on_main_net": false, "idempotency_key": "8d2996cc-f6f4-4c91-9eb9-b405829e88ed", "expiration_time": 1753997254905 } </pre>
Response 400	<pre> { "timestamp": "2025-04-21T21:49:25.922-04:00", "status": 400, "error": "Bad Request", "message": "Transacción no encontrada", "path": "/api/v1/transactions/1468c46c-cd35-4ebc-b929-fb8127545fa6s/cancel" } </pre>
Response 500	<pre> { "timestamp": "2025-04-21T21:49:25.922-04:00", "status": 500, "error": "Internal Error", "message": "Error interno", "path": "/api/v1/transactions/1468c46c-cd35-4ebc-b929-fb8127545fa6s/cancel" } </pre>

Note.

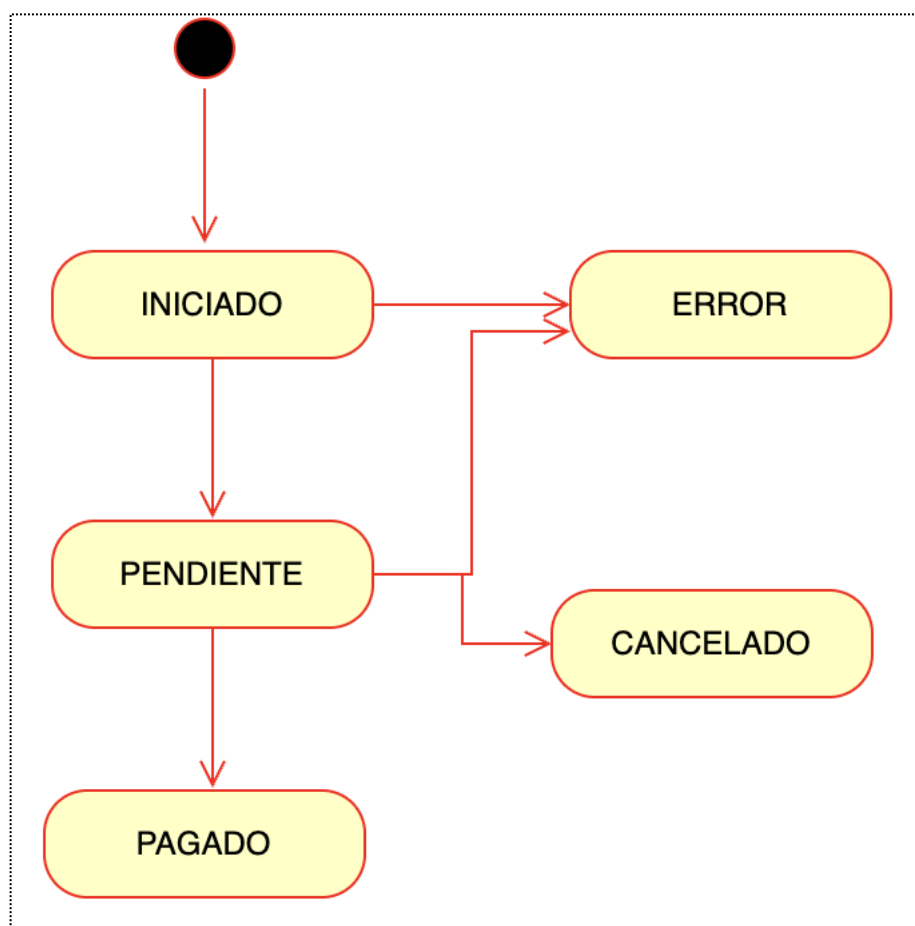
The value of the `created_date` field is the date and time the record was created, expressed in milliseconds.

CHARGE STATEMENTS (TRANSACTION)

The lifecycle of a transaction begins with the INITIATED state. This state, like most others, is only temporary. Stereum Pay has a process called "ReservationMonitor." This process monitors that the order is paid within 10 minutes. If the transaction is not paid within 10 minutes, it will move to the CANCELLED state.

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NAME	DESCRIPTION
INITIATED	Initial state of the transaction
PENDING	Status indicating that the transaction has not yet been paid. If it is not paid, it will be CANCELED.
ERROR	If any error scenario is generated in the job creation process, the position will move to this state.
CANCELED	Status indicating that the charge is CANCELLED, meaning it can no longer be paid.
PAID	Transactions enter this state only when either the Bank or the Blockchain network has confirmed the transaction as successful.



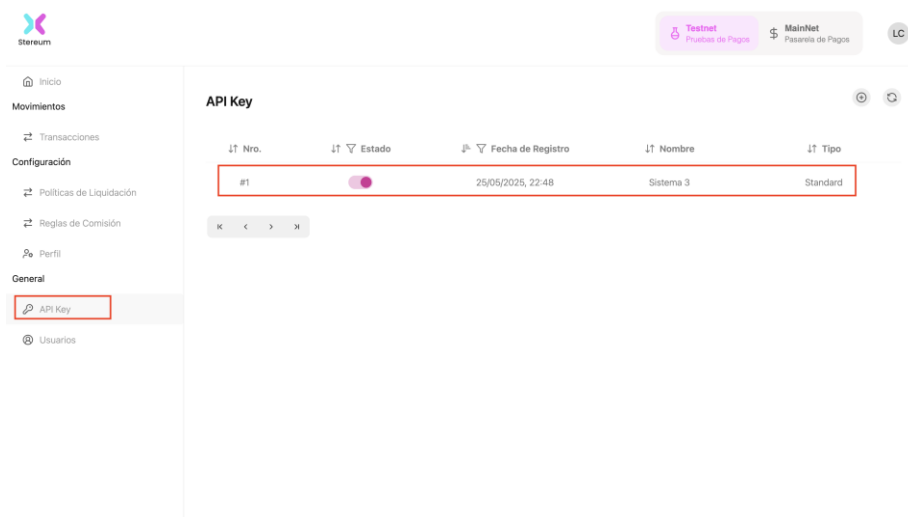
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Figura 4. Árbol de estado

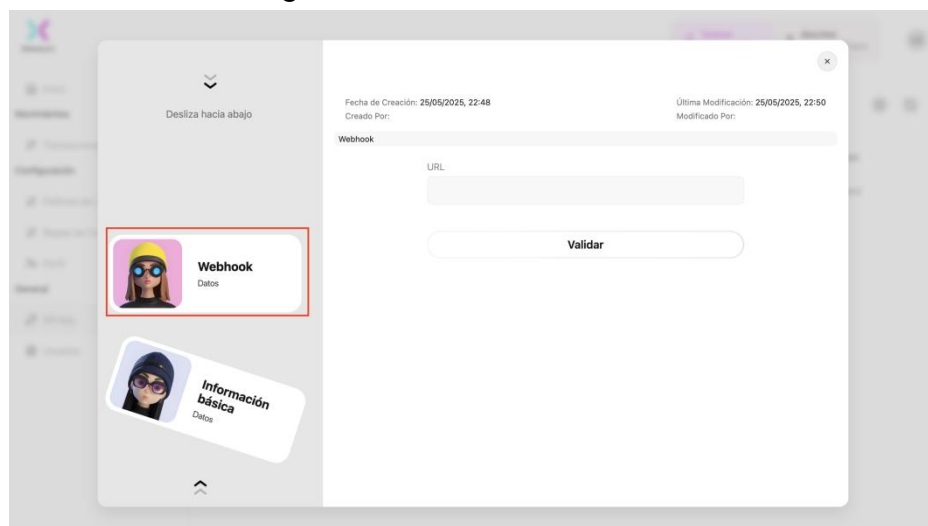
WEBHOOK

Stereum Pay notifies the company's core in real time via Webhook of changes in the charge status (Transaction). To do this, you must configure a Webhook URL for each API KEY in the administration console by following these steps.:

1. Go to the API KEY you created for the integration and click on register.

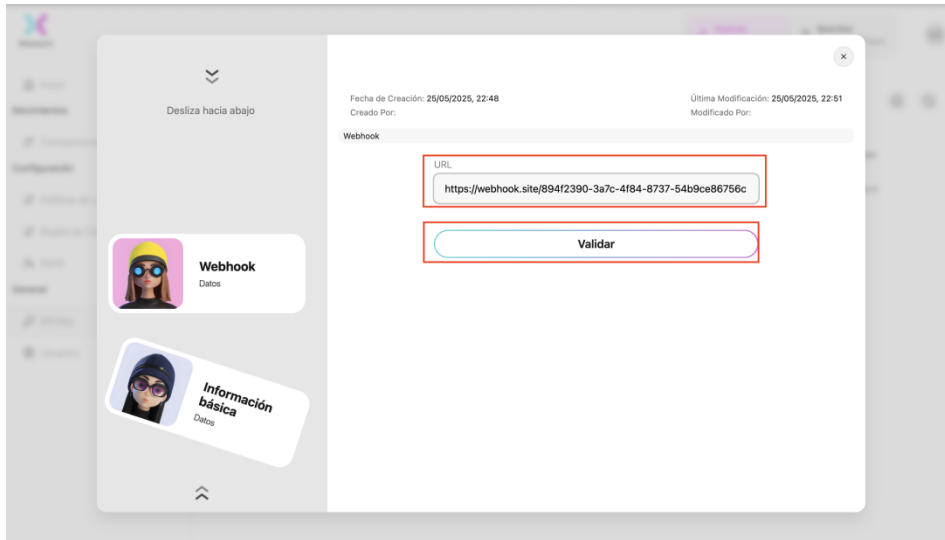


2. Locate the webhook segment.



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3. Enter your webhook URL and press the validate button..



Desliza hacia abajo

Fecha de Creación: 25/05/2025, 22:48
Creado Por:

Última Modificación: 25/05/2025, 22:51
Modificado Por:

Webhook

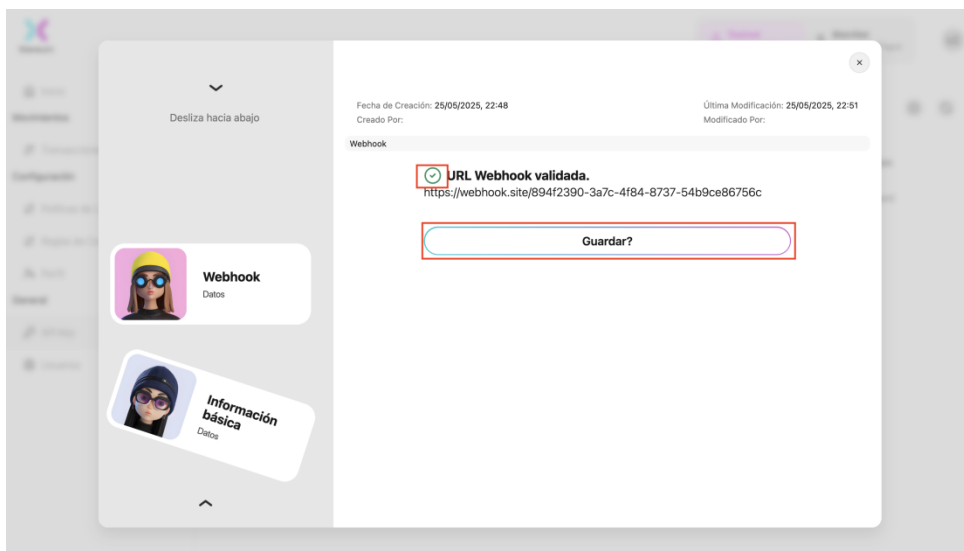
URL
https://webhook.site/894f2390-3a7c-4f84-8737-54b9ce86756c

Validar

Webhook
Datos

Información básica
Datos

4. If you successfully consume the URL, the system will show you the option to save..



Desliza hacia abajo

Fecha de Creación: 25/05/2025, 22:48
Creado Por:

Última Modificación: 25/05/2025, 22:51
Modificado Por:

Webhook

URL Webhook validada.
https://webhook.site/894f2390-3a7c-4f84-8737-54b9ce86756c

Guardar?

Webhook
Datos

Información básica
Datos

In the process of validating the webhook URL, the system will send you the following JSON object via the post method.

```
{
  "notification_type": "test",
  "notification": {
    "hello": "world"
  }
},
```

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```

    "id": "352c7468-f040-4c50-b274-de0475b13e96",
    "timestamp": 1748227866831
  }

```

Just like any notification, it will be sent with all the security measures explained in the Webhook Security section.

Something to keep in mind in the test JSON object is that the notification type has the value "test." This indicates that it is a test notification. Therefore, in the implementation logic, it is recommended to check if the notification type has the value "test." If so, the flow should be interrupted and any JSON object with an HTTP 200 code should be returned.

Example pseudocode

```

Public string webhookStereum(String request){
  x_signature = obtenerCabecera("x-signature");
  x_timestamp = obtenerCabecera("x-timestamp");

  if(!validarFirma(request, x_signature)) {
    Retorna 403;
  }

  if(validarTiempoValidez(timestamp) {
    Retorna 403;
  }

  If(request.notificationType == "test") {
    Retorna cualquierObjetoJson();
  }
  If(request.notificationType == "transaction") {
    ProcesarNotificacion(request);
    Retornar cualquierObjetoJson();
  }
}

```

JSON structure for webhook notification.

```

{
  "notification_type": "transaction",
  "id": "7da7c587-f7d6-46fc-8422-461b4224c993",
  "transaction": {
    "country": "BO",
    "amount": 1,

```

:: <https://stereum.tech>

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```

"amount_received": 1,
"status_description": "Pagado",
"on_main_net": false,
"fee": 0.01,
"idempotency_key": "f64be993-0717-4059-8e8f-05cfb538034a",
"currency": "USDT",
"network": "POLYGON",
"id": "8f26e29d-8a93-4498-90ec-fcbc7814e7ce",
"created_date": 1746884679180,
"payment_date": 1746884689193,
"status": "PAGADO"
},
"timestamp": 1746884689214
}

```

Stereum Pay will send a POST to the Webhook configured with the API Key every time the transaction status changes.

Important note for payments with USDT or USDC!

There is a possibility that the customer pays a different amount than the one requested. In this case, Stereum Pay will verify if the difference is equal to or less than 0.05. If so, the payment will be accepted and the charge will be set to PAID. Otherwise, a manual reconciliation will be necessary according to the procedure explained by Stereum..

WEBHOOK SECURITY

To ensure that the message cannot be intercepted and modified, Stereum Pay signs the JSON with the **HmacSHA256** algorithm and the API KEY as the key each time it sends a POST. The signature result is sent in the request header as **x-signature**. It also sends the current time converted to seconds in the x-timestamp header..

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The resulting signature will always be unique for each request, so the Core System when it receives the JSON of the request, must generate its own signature with the API KEY and compare the generated signature with the one received in the x-signature header, only if both signatures are equal the notification should be processed, otherwise, the http code 403 (Forbidden) should be returned.

Likewise, the time elapsed from when Stereum Pay sent the request until it was received should be calculated based on the x-timestamp header. If more than two minutes have passed, it is recommended to return a 403 (Forbidden) error code.